

Fastronomic

A meal replacement is a substitute for solid food for which the nutrients and calories per portion are carefully controlled. The types of substitutes are soups, protein bars or drinks. Fastronomic are a company that specialise in meal replacement products, with their main product being a protein powder for making a shake. As a business analytics expert, you have been brought in by Fastronomic to help with modelling, coding and planning their production. You will provide them with your code and report on the findings of your analyses.

The company's main product is the meal replacement powder **Fastronomic Alpha**. This product was developed to support people on weight-loss programmes by maintaining a balanced intake of nutrients. It is made using a blend of four powders (**B-base**, **P-protein**, **C-carb**, **O-oil**) that each have differing nutritional properties. Fastronomic are providing the data and model they use to optimise the blend for the Alpha product. They have been using Excel for modelling and optimisation, but as their business grows they wish to use more sophisticated tools that will be efficient and more easily generalisable. They would like you to implement their model in Python and compare the resulting blend to their current mix.

As part of the company's growth, they are introducing a new powder that is high in protein and low in carbohydrate called **Fastronomic Omega**. Market research has identified a market for this product that includes athletes following high protein diets that support muscle development. The research suggests that this is a different market from the Alpha product and its introduction should not cannibalise from the Alpha market significantly. Due to the high protein content of Fastronomic Omega, a new powder has been sourced that has a very high protein content, called **P+-protein**, in order to produce this blend. It is also considerably more expensive than the standard P-protein powder. Fastronomic need you to develop a new blend from the five powders that meets the nutritional requirements of the Omega product at the lowest cost.

Finally, Fastronomic need support in planning for the production of the two products to meet anticipated demands. They would like you to create a model that will allow them to develop production plans for their products. They would also like to highlight any opportunities where it will be cost efficient to expand production by increasing the availability of the ingredient powders.

Data and product blend model

Fastronomic powders are sold in 2kg bags that provides 16 meals (125g portions). A 125g portion of the powder product is a meal substitute that provide a balanced amount of an adults' daily Reference Intake (RI). In Table 1 below we provide the nutritional information about adult RI, Alpha and Omega powders and the five ingredient powders.

Product:	Energy (kcal)	Fat (g)	Carbohydrate (g)	Protein (g)	Salt (g)	Fibre (g)
RI (daily)	2000	70	260	50	6	30
Alpha (125g)	500	12	40	25	0.6	9
Omega (125g)	500	25	10	50	1	10
B-base (100g)	400	11	40	11	0.5	6.6
P-protein (100g)	300	1	10	50	0.5	10
C-carb (100g)	500	0.1	60	0.1	0.5	5
O-oil (100g)	600	65	0.1	0.1	0.2	5.1
P+-protein (100g)	300	1	1	70	1.2	10

Table 1: Nutritional data on daily adult RI, Alpha and Omega products per 125g portion and the powder ingredients per 100g of ingredient.

The cost of the ingredient powders is presented in Table 2, with the selling price of the blended products in Table 5.

Powder:	B-base	P-protein	C-carb	O-oil	P+-protein
Price per kg	£7.00	£8.00	£4.00	£6.00	£12.00

Table 2: Price per kg of the ingredient powers.

The model used by Fastronomic to determine the blend of powders for the Alpha product is provided below. All ingredient measurements are in grams (g) and ingredient prices are in pounds (£s) per gram. The objective is to minimise the ingredient cost per 125g portion, while ensuring the nutritional balance of the product is met. B, P, C, O are used to denote the amount (in grams) of B-base, P-protein, C-carb and O-oil ingredient powders, respectively.

minimise $0.007B + 0.008P + 0.004C + 0.006O$

subject to

$$\begin{array}{rclcl}
 4B & + & 3P & + & 5C & + & 6O & \geq & 500 & \text{(Energy)} \\
 0.11B & + & 0.01P & + & 0.001C & + & 0.65O & \geq & 12 & \text{(Fat)} \\
 0.4B & + & 0.1P & + & 0.6C & + & 0.001O & \geq & 40 & \text{(Carbohydrate)} \\
 0.11B & + & 0.5P & + & 0.001C & + & 0.001O & \geq & 25 & \text{(Protein)} \\
 0.005B & + & 0.005P & + & 0.005C & + & 0.002O & \geq & 0.6 & \text{(Salt)} \\
 0.066B & + & 0.1P & + & 0.05C & + & 0.051O & \geq & 9 & \text{(Fibre)} \\
 B & + & P & + & C & + & O & = & 125 & \text{(125g serving)} \\
 B & & & & & & & \geq & 0 & \text{(Non-negativity constraints)} \\
 & & P & & & & & \geq & 0 & \\
 & & & & C & & & \geq & 0 & \\
 & & & & & & O & \geq & 0 &
 \end{array}$$

Fastronomic's analysis requirements (60 Marks)

Analysis requirement (Alpha blend): Fastronomic would like you to implement their Fastronomic Alpha model in Python. (Ideally, all of the analyses for Fastronomic should be conducted in Python.) You should compare your results from Python to the blend produced by the company who used Excel for their analyses. They expect any differences between the two blends to be minor. Table 3 presents the amount (g) of each ingredient powder used to make a 125g portion of Fastronomic Alpha, rounded to 2 decimal places. Table 4 presents the nutritional information resulting from the blend.

Powder:	B-base	P-protein	C-carb	O-oil
g per 125g portion	58.81	37.00	21.28	7.91

Table 3: Amount (g) of the ingredient powders using in a 125g portion of Fastronomic Alpha.

Product:	Energy (kcal)	Fat (g)	Carbohydrate (g)	Protein (g)	Salt (g)	Fibre (g)
Alpha (125g)	500.09	12.00	40.00	25.00	0.60	9.05

Table 4: Nutritional information of the resulting Fastronomic Alpha blend.

Note that, when solving the model, the nutritional contribution may slightly exceed the target value. Similarly, when rounding the amount of ingredient powders used to 2 decimal places and then calculating the nutritional information using these rounded weights, as in Table 4, the nutritional contribution may be slight below or above the target value. Small discrepancies such as this are expected and are normally within allowable tolerances for production of the product.

Analysis requirement (Omega blend): With the launch of Fastronomic Omega, the company would like you to model and optimise the ingredient powder blend that will minimise the ingredient cost per 125g portion, while ensuring the nutritional balance of the product is met. Here you will need to use the P+-protein powder along with the four ingredient powders used for the Alpha blend. When reporting the resulting blend for Fastronomic Omega, as for the Alpha blend, round the weights of the ingredient powders to 2 decimal places then calculate the nutritional information for the Omega product using these rounded weights.

Analysis requirement (Production planning cycle): Fastronomic need a model to support production planning. The objective of the model is to determine how many 2kg packs of Fastronomic Alpha and

Omega to produce in order to maximise their profit. Here, profit is the difference between revenue generated from selling the products and the cost of ingredients used in production.

Each 2kg pack of Alpha and Omega product will use the blend of powders identified in the Alpha and Omega requirement analyses. From these analyses you will also be able to determine the ingredient production costs of a 2kg pack of either product. In Table 5 the selling price of 2kg packs of the Alpha and Omega products is provided.

Product:	Alpha	Omega
Selling price of 2kg pack	£25.00	£30.00

Table 5: Selling price of 2kg Fastronomic Alpha and Omega packs.

Table 6 provides the maximum amount of ingredient powders that the company are able to source for this production cycle at the prices given in Table 2. Note that, the company only purchase the amounts required for the production plan this cycle as given by the resulting solution to the model.

Powder:	B-base	P-protein	C-carb	O-oil	P+-protein
Availability (kg)	30,000	20,000	20,000	10,000	15,000

Table 6: Availability (in kg) of the ingredient powders used to make the Alpha and Omega blends.

For the first analysis using your model, Fastronomic would like to evaluate what the optimal production plan is. The company are aware that it is possible to increase production this cycle if they are able to source additional ingredient, albeit at a higher purchase cost per kg. They would also like you identify which ingredients powders that if they purchase more of will allow them to increase production levels. If they were to select one of these ingredient powders to purchase more of: which ingredient powder should they choose? What price should they be willing to pay per kg for the additional ingredient? How much can or should they purchase? Finally, what will the optimal production plan and potential profit be with the additional ingredient powder?

Hint: You can get sensitivity analysis results if your decision variable are continuous variables. However, Fastronomic are producing integer amounts of packs. You could solve the problem using continuous decision variables then convert to an integer solution, or resolve the problem as an integer program once you have obtained the information from the sensitivity analysis results. Make sure you report how you did this in your report if you attempt to provide a solution to this requirement.

However, Fastronomic are cautious about the demand for the new Omega product. Based on historical demand and market research projections as appropriate, Table 7 contains data on the anticipated demand for the next production cycle. Assuming that additional ingredient powders will not be purchased this production cycle, determine the optimal production plan that considers the demand projections and the optimal profit under this plan. Compare this plan to the plan that does not take into account of the demand information and report on your findings.

Product:	Alpha	Omega
Demand (in 2kg packs)	At least 27,000	Up to 8,000

Table 7: Demand estimates (in 2kg packs) for the Alpha and Omega products for the next production cycle.

Management Report (20 Marks)

The maximum length of the report is 10 pages (not counting the title page or the appendix). You can use any readable font (i.e., not too small) and use single spacing. There is no penalty for a shorter report, although it is unlikely that you would be able to address all the issues properly in a very short report. You need to provide the information requested and explain the analysis you have done.

The report should follow the guidelines set out in the recorded lecture on Report Writing. However, since this is a short report:

- You do not need to start different sections of the main body on separate pages;
- A contents page is not required. If you include a contents page it will count towards the total page limit;
- For the appendix, state which analyses tasks have been completed in Python. You then need to copy and paste the relevant Python code (as a text, **not screenshot**) into the appendix.
- No other details should be entered in the appendix. Anything relevant to the report should be in the main body of the report. Any additional details in the appendix that have not been requested will be ignored and will not contribute to your overall mark.

Python Code (20 Marks)

These marks will go to programs that are:

- Well-structured and well-commented
- Intuitive to use (i.e., straightforward for us to run the code)
- Generalisable (i.e., your code can be used with any parameter values)
- Flexible (i.e., provide user options via a user interface, covering all the analysis performed)

An example of a user interface is as follows:

```
> Choose one of the following options:
> 1: The amount of each ingredient powder used to make a portion of Alpha
> 2: Nutritional information of the resulting Fastronomic Alpha blend
> . . .
> Q: Quit the program
1
> B-base: 58.81, P-protein: 37.00, C-carb: 21.28, O-oil: 7.91
> Do you want to continue (Y/N)?
Y
> Choose one of the following options:
> 1: The amount of each ingredient powder used to make a portion of Alpha
> 2: Nutritional information of the resulting Fastronomic Alpha blend
> . . .
> Q: Quit the program
2
> Energy(kcal): 500.09, Fat(g): 12.00, Carbohydrate(g): 40.00, Protein(g):
25.00, Salt(g): 0.60 Fibre(g): 9.05
> Do you want to continue (Y/N)?
N
```

The program should run without errors and sensibly uses programming techniques developed on the module. The source code should be programmed with Jupyter Notebook as an Editor. It is important to break your code into small, meaningful functions and appropriately comment your code.

Note: If you are unable to use Python for all of the analyses, you may use Excel in order to gain marks for that element of the coursework instead. However, your mark for the Python code will be negatively affected.

Academic Malpractice

This coursework is to be done as an individual piece of work. All of your analyses and the whole of your report must be your own work alone. Collusion between students and plagiarism are regarded as very serious offences and will be penalised severely. Please note that software to detect plagiarism and collusion will be used.

Submission Instructions

- Your coursework should be submitted to the MSCI151 Moodle site as two documents. Your python code and a Word or PDF document containing the management report.
- If possible, please submit your report as a PDF file (you should be able to save it in PDF format from Word). Alternatively, submit it as a Word file if that is not possible.
- Your python code should be submitted as a single Python file (.ipynb) that contains the Python code used to create the output/results for the analysis.
- The name of the Python file should be your Library Card/Student Number. E.g., 12345678.ipynb, where 12345678 is an example Library Card/Student Number.
- The place to submit your files is in the MSCI151 Moodle site in the Methex 2 (individual) submission link under the “Individual Methex (Summer term)” section.
- You are required to provide both the Python file **and** the relevant Python code segments in the appendix to the report.
- Do not submit any other files.
- The deadline for submission is 4pm on Thursday 11th May 2023.

Remember to give yourself the opportunity to complete the task with time to spare. Leaving it to the last minute often results in a poor submission and a low mark. Late work without an agreed extension will be penalised according to the University rules (deduction of marks as shown in the assessment rules table for work up to 3 days late; a mark of 0 for work more than 3 days late). Extensions may, in certain rare circumstances, be given but to obtain an extension you must download the extension request form from the Management Science FAQ Moodle site. You are very strongly advised to make back-up copies of your work during the project, e.g. on your hard drive, a USB and/or on a cloud storage service – **loss of work due to computer problems will not be accepted as a reason for an extension.**